

CMPE452 Assignment Part A – Controlled Experiments, Best Configuration, and Reporting Guidelines

Dear students,

This is a reminder regarding the design and reporting of your hyperparameter experiments for the **CMPE452 Assignment Part A**.

Controlled Experiment Approach:

To meaningfully and reliably analyze the effect of each hyperparameter on model performance, all experiments must follow a controlled experiment methodology.

This means that in each experiment, only one hyperparameter should be changed, while all other parameters remain fixed. This allows you to clearly isolate and observe the individual effect of each parameter.

Hyperparameters to Be Tested:

You are expected to perform experiments on the following hyperparameters:

- conv_kernel_size
- conv_dropout
- activation_fn
- batch_size
- optimizer_type
- num_conv_layers
- fc_hidden_units
- pool_kernel_size
- pool_stride
- fc_dropout
- num_epochs

Experiment Rules:

- For **activation_fn** and **optimizer_type**, testing up to **3 different** values is sufficient.
 - activation_fn: ReLU, Tanh, Sigmoid
 - optimizer_type: adam, sgd, rmsprop
- For all other hyperparameters, you must test **at least 6** different values.
- Each experiment must be isolated — you should modify **only one parameter at a time**, keeping the rest fixed.

Additional Note on `num\_epochs`:

You are also expected to include `num\_epochs` as one of the tested parameters in your experiments. This parameter should be evaluated with at least 3 different values, such as 5, 10, and 20 (or more). It is important to leave meaningful intervals between the values (e.g., avoid using values like 9, 10, 11) in order to observe clear trends in model performance.

Creating the Best Configuration:

Once you have completed all controlled experiments, you should select the best-performing value (i.e., the one with the highest validation accuracy) for each hyperparameter.

These selected values should be combined into a single configuration, referred to as the **best configuration**.

This configuration must be used to train a final model, and only this model should be evaluated on the test set.

⚠ The test set must not be used during hyperparameter tuning. It should only be used once, after finalizing the best configuration.

During testing, you must report key metrics including:

- Test accuracy
- Confusion matrix
- Classification report (precision, recall, F1-score)

Reporting Requirements:

Your project report must include detailed tables and plots of your results.

Specifically, for each hyperparameter tested with 6 different values, you are required to include the following:

- Two separate graphs:
 - One showing validation accuracy vs. hyperparameter values
 - One showing validation loss vs. hyperparameter values
- A summary table comparing all tested values and their corresponding performance
- A brief discussion/commentary interpreting the results:
 - Which value performed best?
 - Why might that value have led to better performance?

These comparisons are essential to demonstrate your understanding of how each hyperparameter influences model behavior and to justify the choices made in your best configuration.